KAPASA MAKASA UNIVERSITY

DEPARTMENT OF AGRICULTURAL AND AQUATIC SCIENCES

BACHELOR OF SCIENCE IN AGRO-FORESTRY

BACHELOR OF SCIENCE IN SUSTAINABLE AGRICULTURE

2024 SESSIONAL EXAMINATION

COURSE NAME: PLANT PROTECTION

COURSE CODE: (NR 430, SA350)

TIME ALLOWED: 3 HOURS

DATE: 17TH OCTOBER, 2024

INSTRUCTIONS

i. This question paper comprises of seven **(7)** questions

ii. You are required to ATTEMPT a total of **five (5)** questions

iii. Each question carries **20 marks**

iv. Question 1 in SECTION A is COMPULSORY

v. Choose 2 questions from SECTION B

vi. Choose 2 questions from SECTION C

vii. Clearly, write your Student Identity Number on the Answer Booklet provided

viii. Write neatly and orderly to avoid loss of marks

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- diptere (flies)
- hymenoptere (bee)
- orthoptere (grasshopper)

SECTION A

QUESTION 1 (compulsory)

(a) Insect taxonomy is a very important branch of science to agriculturalists as it deals with recognizing, describing and naming of species, and classification of such insects into a ranked system. State and in details describe seven (7) "*Orders*" into which insects can be classified including examples of insects that belong to such "*Orders*". **(14 marks)**

(b) In details, explain how bacteria reproduce and cause plant diseases. **(6 marks)**

SECTION B

QUESTION 2

(a) Explain how plants defend themselves against infections using pre-existing defense structures **(12 marks)**

(b) Define the following terms

i. Stridulation

ii. Phytopathogenesis

iii. Conidia

iv. Phylogenetics **(8 marks)**

QUESTION 3

(a) Explain how *Orientation and Recognition* of hosts takes place as a behavioural activity of insects before they establish themselves on the host plant **(10 marks)**

(b) Nematodes are known to cause a number of problems and diseases in plants. Clearly, explain the damages caused by nematodes and how they cause such damages **(10 marks)**

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QUESTION 4

(a) In details, discuss plant disease tetrahedron **(12 marks)**

(b) List various symptoms and signs caused by plant bacterial diseases **(4 marks)**

(c) Explain in details how viruses invade the plant and cause diseases **(4 marks)**

SECTION C

QUESTION 5

(a) Integrated Pest and Disease Management is based on six principles of plant disease management and it integrates multidisciplinary approaches for the management of plant diseases. Discuss **(12 marks)**

(b) Briefly, explain how Polymerase Chain Reaction (PCR) works as a plant disease identification method **(8 marks)**

QUESTION 6

(a) Weeds are known to cause tremendous damage in crop fields if not controlled. Discuss various levels of weed classification **(12marks)**

annual
bi- annual
Perinial

(b) Briefly explain various characteristics of arthropods **(8 marks)**

QUESTION 7

(a) Explain various cultural practices that aim at reducing the adverse effects of plant diseases **(10 marks)**

(b) Biological control is among the methods used in weed management and control. Clearly, discuss the merits and demerits of this method. **(10 marks)**

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